

PURBANCHAL UNIVERSITY

2026

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

MCA103/MCA113: Operating System (New/Old Course)

Candidates are required to give their answers in their own words as far as practicable.

Group A

Answer TWO questions.

2×12=24

1. For the processes listed in the following table, create a Gantt chart illustrating their execution and compute the average waiting time and average turnaround time using:

- (a) First Come First Serve (FCFS)
- (b) Shortest Remaining Time Next (SRTN)
- (c) Priority Scheduling (Preemptive)
- (d) Round Robin (Quantum =2 ms)

Process	Arrival Time (ms)	Burst Time (ms)	Priority
P1	0	6	3
P2	1	8	1
P3	5	7	2
P4	2	3	2

2. Differentiate paging and segmentation with suitable example. Compute the total page faults for the given reference string with 3 page frames using LRU, FIFO and optimal page replacement algorithms.
Reference string: 3, 2, 0, 2, 5, 3, 5, 2, 3, 0, 5, 2, 0, 5, 2, 3
3. What do you mean by client-server computing and how it is different from distributed computing? Explain the design issues in distributed operating system. 5+7

Group B

Answer SIX questions.

6×6=36

4. Assume we have disk with 150 tracks and currently head is at track number 38. What will be the total disk arm movements for the algorithms SCAN, C-SCAN and LOOK for processing IO requests queue: 53, 65, 22, 11, 45, 85, 19, 77, 90, 8? **Contd. ...**

(2)

5. What is passive attack? Differentiate between authorization and authentication. .
6. What is producer consumer problem? Solve using semaphore concept. .
7. Explain the types of memory partitioning with their merits and demerits. Give suitable illustrations. .
8. What is Ostrich algorithm? Explain deadlock prevention and recovery in detail.
9. What are file attributes? Explain the file allocation methods. .
10. Write short notes on (Any TWO):
 - (a) Interprocess communication .
 - (b) Structure of operating system .
 - (c) Types of distributed system